

KILNDECK · DESIGN DECISIONS REGISTER

KilnDeck design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kilndeck		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Register opened; all decisions made under Amish's 2026-10-03 pre-approval

KilnDeck design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list decisions.

■ Open decisions

None. All decisions were made under Amish's 2026-10-03 pre-approval.

■ To confirm when parts are bought

These are facts that can only be settled with real parts or a real kiln. None changes a decision; each may change a size.

Table 1. Items to confirm.

#	What to confirm	Why it matters	Source
1	Survey of the partner kiln: trench width, wall-top width, level and condition, crust level and temperature, feed hole grid	Sets the track gauge (now 6,600 mm), the deck length, the spout reach and the floor temperature figures	KND-CAL-001, section 1
2	Bearing limit of the kiln wall tops from the survey	R8 is met only against the interim 0.20 MPa limit	KND-DDR-002, A2
3	Track wheels: rating of at least 500 kg, groove widths 24 and 44 mm, axle length for the drive sprocket	Worst wheel load about 447 kg; the inner wheels must float 12 mm	KND-CAL-001, section 9
4	Calcium silicate board: compressive strength at least 0.6 MPa and density near 240 kg/m ³	A heel puts about 0.41 MPa on the board; the panel mass assumes this density	KND-CAL-001, section 8
5	Wall thickness and straightness of the SHS bought for the trusses	The truss modules have under 1 kg of margin on the 40 kg limit	KND-CAL-001, section 5
6	Hand wheel rim: 12 holes or room to drill them on a 298 mm circle	The lock pin and the 24 mm stopping step depend on it	KND-CAL-001, section 10
7	Bulk density and lump size of the partner kiln's coal and biomass	Sets the dose per pocket and the grille gap	KND-CAL-001, section 4
8	Silicone grip sleeve to fit a 40 mm square rail, rated to 200 °C	Grip and heat protection on the top rails	BOM line 16

■ Value engineering

Value-engineering target: USD 3,500 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 2,627 (USD 873 under the target). Main cost drivers and savings worth trying:

- The largest lines are the eight floor panels (USD 544, of which about USD 200 is insulation board), the eight truss modules (USD 432), the track for two 12 m runs (USD 368 plus USD 91 of pads and USD 48 of stops), the travel drive (USD 248) and fasteners, paint and consumables (USD 180).

- The track is about a fifth of the cost and is reused as the fire moves, so a longer working run (the fire moves 6 to 10 m a day) would add about USD 51 per 3 m of both tracks.
- Savings worth trying: aluminium tread sheet in place of steel on the floor panels (lighter, but dearer per panel); reclaimed steel channel for the track; a single chain from the hand wheel direct to the outer-end drive if the line shaft can be shortened after the survey.

■ Decisions made

Table 2. Decisions made.

Date	Decision	Decided by	Record
2026-10-03	TRL 2 review items D1 to D9: deck spans the trench on two wall-top tracks; movable track on the straight sides; hopper carriage beside the deck; hand wheel and line shaft; 1.1 m guardrail with knee rail and toe board; insulated floor; coal and biomass; first co-design candidates (a clean-kiln programme in Bangladesh, then a kiln owners' association in Punjab, Pakistan, not yet approached); budget, pitch and problem unchanged	Amish, pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost."	KND-DDR-001
2026-10-03	Design for construction, changes C1 to C15: track modules on pads with sliding fishplates, end stops, end trucks with derailment guards, guiding and floating wheels, bolted truss modules, U-frame bearers, insulated floor panels, inward-opening gate, travel drive with lock pin, carriage rail and carriage, hopper grille, rotor with crank over the rail, swing spout, grip sleeves, assembly over the cold zone	Amish, same pre-approval	KND-DDR-002
2026-10-03	Wheels rated at least 500 kg; interim wall bearing limit 0.20 MPa until the survey (conservative; relaxed only by a survey); no shade canopy on the first prototype, with R7 reported as not met in full sun and heat-resistant footwear kept (relaxed only by a canopy wind check and logged floor temperatures); curved kiln ends not served	Amish, same pre-approval	KND-DDR-002, A1 to A4
2026-10-03	Appearance model departures: a bolt pattern, a nameplate, a load label and a kiln wall and crust strip drawn for the renders only	Amish, same pre-approval	docs/REVIEW.md, TRL 3 section